

# CP #6

William Dong

2026-05-05

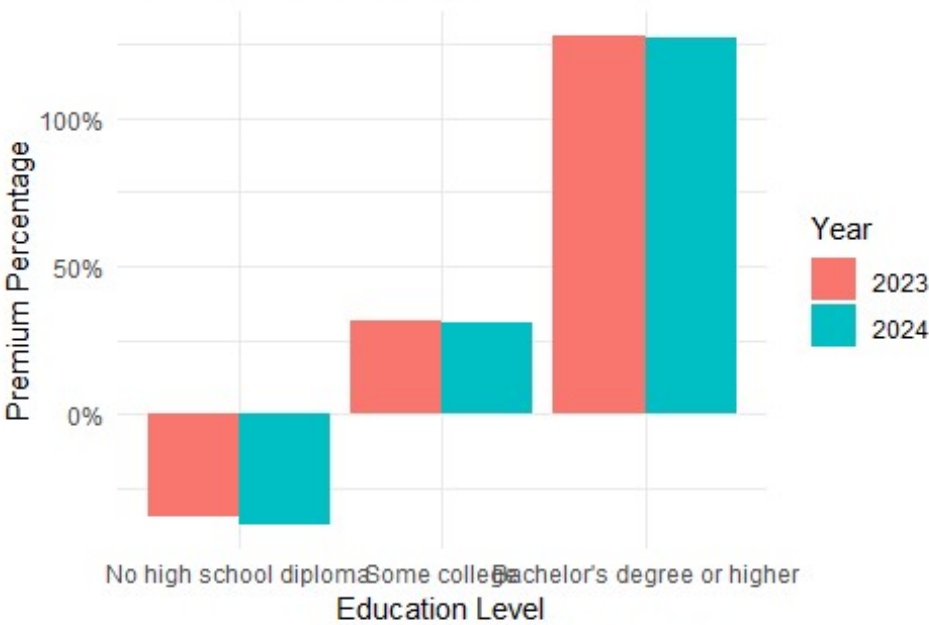
The education results show a clear earnings premium associated with higher educational attainment. Households headed by individuals with a bachelor's degree or higher earned substantially more than all other education groups in both years, reinforcing the strong income advantage of college completion. Those with some college also earned more than high school graduates, though the difference was much smaller.

A notable exception appears among households without a high school diploma. This group was the only education category to experience a decline in real median income from 2023 to 2024, indicating that households with the lowest educational attainment became slightly worse off in inflation-adjusted terms. In contrast, all other education groups saw modest gains, suggesting that income growth in 2024 was concentrated among households with at least a high school education.

--- Analysis of Earnings Premium Relative to High School Graduates ---

|   | education                   | premium_2023 | premium_2024 | premium_delta |
|---|-----------------------------|--------------|--------------|---------------|
| 1 | No high school diploma      | -0.3437555   | -0.3682589   | -0.024503401  |
| 2 | High school graduate        | 0.0000000    | 0.0000000    | 0.000000000   |
| 3 | Some college                | 0.3189520    | 0.3100496    | -0.008902316  |
| 4 | Bachelor's degree or higher | 1.2724891    | 1.2718713    | -0.000617828  |

Higher Education Earnings Premium (2023 vs 2024)  
Relative to High School Graduates



Source: U.S. Census Bureau

---

## CASE 2: GENDER

---

The gender comparison shows that male-householder households earned more than female-householder households in both 2023 and 2024. Although median income remained higher for men, female-householder households experienced income growth between the two years while male-householder households experienced a slight decline.

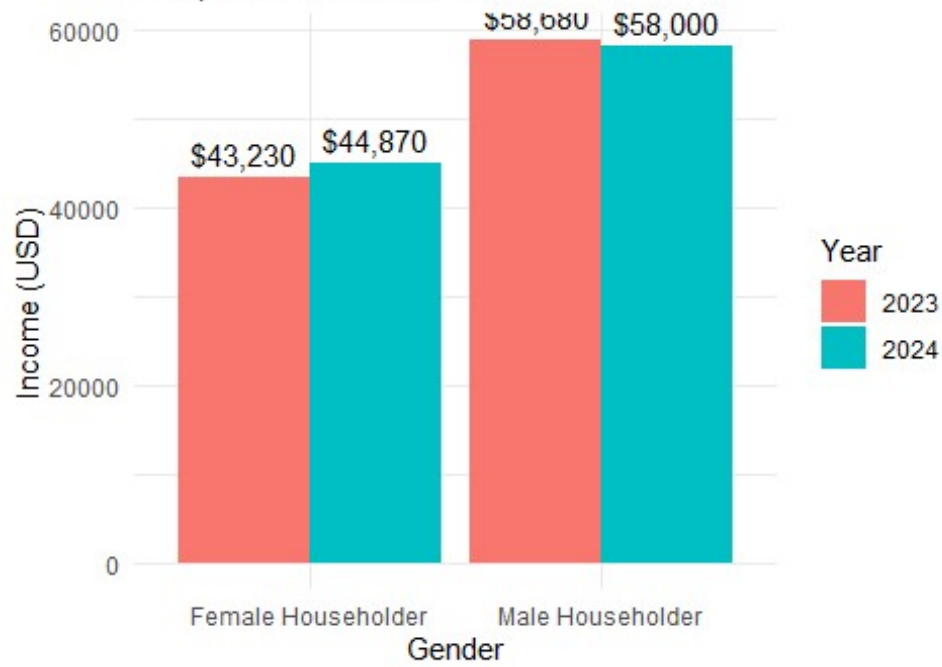
As a result, the female-to-male income ratio improved in 2024, indicating that the gender income gap narrowed slightly. Despite this improvement, the gap remains substantial, with female-householder households continuing to earn considerably less than male-householder households overall.

--- Gender Income Ratio (Female / Male) ---

|   | ratio_2023 | ratio_2024 | improvement | status        |
|---|------------|------------|-------------|---------------|
| 1 | 0.7367076  | 0.7736207  | 0.03691312  | Gap Narrowing |

## Median Income by Gender

Comparison of 2023 vs 2024



---

## CASE 3: AGE

---

Age is also strongly associated with household income. Median household income generally rises with age through prime working years, peaks among households aged 45 to 54, and then declines in older age groups. This pattern reflects the typical life-cycle of earnings, where income increases with labor market experience and career advancement before falling as households approach retirement.

Among all age groups, households aged 45 to 54 were the most financially advantaged, reporting the highest median income. Younger households, especially those aged 15 to 24, earned substantially less, reflecting lower labor force attachment and earlier career stages. Income also declines among households aged 65 and older, which is consistent with retirement and reduced wage income.

Several age groups experienced statistically meaningful changes between 2023 and 2024, particularly those aged 15 to 24, 35 to 44, and 45 to 54. This suggests that income changes in 2024 were not the same as some working-age groups seeing more measurable shifts than others.

```
# A tibble: 1 × 10
  age_group households_2023 median_2023 moe_2023 households_2024
median_2024
  <chr>          <chr>          <chr>          <chr>          <chr>          <chr>
1 65 years and... 37630          56130          1044          39290          56680
# i 4 more variables: moe_2024 <chr>, pct_change <dbl>, pct_change_moe <chr>,
# significant_change <lgl>
```